DECO3800/7380 Design Computing Studio 3 2024
Stephen Viller, School of EECS
Assignment 2-1 : Report Due 25 March 2024, 13:00 AEST
Presentation in week 5 Studio, 21 March 2024
Learning Objectives: 1, 2, 3, 4, 5, 6

Milestone 1: Problem Identification

Problem Identification by @Lars Schoebitz https://flic.kr/p/G9253Q

Presenting your project concept

Introduction

At this stage of the project, your team should have collaboratively discussed and explored a problem domain or challenge to focus on, and have an initial identification of the design direction you are taking. For this first project milestone, you need to communicate the context of the problem you have identified, and the current status of your team's exploration

of the solution space. Given the nature of the problems and challenges we have been focusing on in studio this semester, you will most likely need to narrow your focus to a specific aspect of the problem in order to scope out a more interesting and realistic project.

Given the relatively early stage of the project process, it is inevitable that your understanding of the design challenge and potential solutions at the moment is partial, and likely to change. This first version of your report—the problem identification—needs to present the current 'snapshot' of your thinking, and identify what work the team is planning in order to address any gaps in knowledge, or any partial understandings that currently exist. **Please do not try to "get it right" at this stage**--our expectation is that you will need to revise your project in the light of what you learn along the way. **If you lock yourselves in to a solution too early, you are likely to have problems later**. Remember the processes we have already followed in the Bad Ideas workshop and further discussions around domain explorations. You should be thinking in terms of generating many diverse ideas and then evaluating their relative merits before selecting what you think is your strongest candidate (and justifying why it is a strong candidate).

Task Description

This assessment item is the first checkpoint of your studio project. Following team formation, you will create a running document that will gather together key information about your project. This document will be worked on throughout the semester, and will ultimately be submitted as the final documentation for your team's project. At this stage, the most important aspect of the submission is to identify **what you are planning to do** and **why you have chosen do it**. In other words, what is the specific problem/challenge or design opportunity you have identified, what do you know about it so far, what ideas have you had in the solution space, what are the strengths and weaknesses of your ideas, and which external sources (references, links to existing projects, etc.) have informed your work?

From a wholistic perspective at this stage, your report, and the associated presentation, will need to deliver:

- A discussion which explains the problem context you have identified and the background research and exploration you have conducted so far into the problem and solution spaces
- Decisions you have made about the potential design direction, including alternatives considered and justifications for your choices
- Information regarding the relative merits of your potential solutions. This might be in terms of costs, ethical considerations, security and other risks and benefits to the stakeholders.
- An initial/high level plan and breakdown of the work required for this semester's design activity

- A discussion of the risks to the overall project's success, and some techniques for mitigating these. These should be thought of in terms of both content (what you are focused on) and process (what you will do and how you will work as a team to get it done)
- Early prototyping[1] work on selected parts of the problem

These will be presented in the form of

- A report to be submitted via blackboard to receive formative feedback
- A 10-minute pitch to be presented to your Week 5 studio session (5 minute presentation and 5 minutes for questions)

In keeping with the nature of assessment for Studio 3, the criteria set out here are intended to provide you with support for what your report and presentation should focus on, and are not intended as a detailed specification for what you should prepare.

Report

The end point for your project this semester will be a *Proposal* for a solution to the problem or opportunity you have identified, so as a team you should be undertaking activities that lead to a better understanding of *what* you are designing and *why*. The essential design challenge for your team, therefore, is to develop a well informed and justified set of requirements for a solution to the problem that you have identified.

Required content

Your report will be the first iteration of what will evolve into your final proposal documentation at the end of the semester. In this first iteration, the main focus should be on establishing what your project is about in terms of the challenge or problem space you are addressing, the specific problem you are focused on, your current ideas for the designed solution, the background research and reading that you have done so far, and what this means for the remainder of the project. You should ensure that this version of your report covers the following content. Due to the diverse nature of projects across the whole of Studio 3, each team will need to decide how the following applies to their project.

- Overview
  - The overall aims and motivation for the project, the challenge/problem space you have selected, why it is important
- Background reading/literature review
  - What existing work have you found in relation to your project? This might be published academic research in peer-reviewed journals and conferences, academic or industry-

[1] This is explicitly **not** asking you to build a prototype of the final envisaged product. Prototyping in this context describes the design mock-ups and activities you undertake in order to try out aspects of your design and learn more about the problem/solution by testing them with people.

based projects also addressing the problem, news or magazine articles from reputable publishers, or even blog posts from recognised experts in the field. What have you done to consider how the different items of background research relate to each other and combine to influence and/or inspire what you are proposing?

- Stakeholders
  - Who are the key people that you need to engage with/design for? Who are or will be impacted by the project? This could be about the people or organisations who would use the proposed solution, but there could be other significant groups who could benefit from (or be harmed by) the success of your proposed solution
- Ethical considerations
  - When we are designing with and for people, there will always be ethical concerns that need to be addressed in the project. These are not a list of items that you will need to think about later once you have completed the design. They should be considered up-front and ideally be used to help determine the direction of your project. Your presentation and report should include a section that explains what your team has so far identified as ethical concerns, and what you have so far done to address them.
- Initial Conceptual model
  - What is your current understanding of the solution space? This will necessarily be incomplete/sketchy. The idea is to communicate your current ideas about the design. The approach used in HCI conceptual models will probably work for most projects, but other approaches are fine where there is a better fit for the specific project. Remember you are doing this to get feedback on your current design ideas rather than to give the "correct" answer.
- Questions/areas of investigation
  - Where are the gaps in your knowledge? You can't possibly know everything that you need to know about the problem or the solution yet. What have you identified as questions that you need answers to in order to make progress with your project? What activities do you need to plan to find out the answers?
- User research/prototyping/sketching/mock-ups
  - What have you done so far to gather data to support your design direction (and to answer any questions identified)? This might be the result of interviews with stakeholders or domain experts, observations of significant activities or locations, low fidelity prototypes or sketches produced to elicit feedback from stakeholders, technology spikes to explore the feasibility of particular solutions, etc. (This is not a list of what every project should have covered, it is a list of possibilities to adapt to your project where applicable). Remember, this studio is **not** about producing a first version

of the solution. The prototyping you will do this semester will be all related to improving your knowledge about the problem/solution space.

- Project plan
  - How will you answer the questions you have identified and iterate on the conceptual model? What work/activities do you need to undertake between now and the end of semester in order to arrive at a final proposal for your project? What risks have you identified that might threaten the success of the project? The plan at this stage will not be detailed or final and will certainly require further refinement, but it should be *specific to your project*. Consider interdependencies between tasks, iteration, etc.

Your task for the rest of the semester will be to understand the problem you are solving in more detail, leading to the identification of a proposed solution. The documentation will grow and evolve as you explore the problem space, identify aspects requiring further investigation, conduct studies to answer questions identified, document the outcomes, and iterate the process.

A report template is provided in the assignment folder on Blackboard if you wish to use it as a starting point, but this is a resource for you rather than a requirement for it to be used. There is no word count for this.

Presentation

Presentations will take place in class in parallel sessions. The presentations are an opportunity for each team to practice communicating about your project to a wider audience, and to receive feedback on your work so far. It is up to each team to decide how to incorporate any feedback received into your report.

There will be a strict five minute time slot for each presentation, followed by five minutes for questions. If necessary, we will make adjustments for technical problems during the presentations, but there will also be a need to get through all of the presentations for each session so there will be practical limits on this. Presentation sessions will be recorded to assist with the feedback process. A sign-up sheet will be published via MS Teams to allow you to sign up to specific time slots.

Criteria

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
"Depth of Treatment" and "Required Content" are weighted higher than "Communication" and "Professionalism" in the overall grade. However, the proposal should be developed to a standard such that it would be suitable to present to a potential investor, where the standard of production and professionalism of the team can influence a decision as strongly as the quality of the idea. Remember, these criteria are about the quality of your work and are deliberately written in a way to allow you flexibility to fit them to your project context, rather than requiring you to change your project to fit them. If you are having problems understanding any criteria in the context of your project, please discuss with your tutor or with the teaching team more broadly.			
DEPTH OF TREATMENT			(weight 5)
Background reading/literature review (*what existing work has informed the project?*)	Of high quality showing evidence of wide research, extending well beyond the starting points identifying the problem or challenge	Adequate, but neither extensive nor exceptional	Shows signs of hurried or inadequate research
Utilisation of Sources (*how well has existing work been used?*)	Sources are cleverly used to support the conceptual and technical work identified in the report	Sources could be better used to support the proposal	Inadequate use of sources to support the concept and/or technology design
Creative work (*how much does the design reflect the team's work rather than reinventing the wheel?*)	Original and creative synthesis of materials	Adequate, but not original use of research materials	Lacks originality. E.g. the project is very close to an exact copy of an existing product.
Relevance to challenge or problem space (*how well does the proposed work address the identified problem?*)	Displays clear understanding of where proposed work fits into the bigger picture of the problem of challenge context	Shows occasional understanding of how the team's work is related to the context	Presents proposed work in isolation, with no connection at all to broader context

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
REQUIRED CONTENT			(weight 5)
Stakeholders (*which people and/or organisations have a stake in the problem being solved?*)	Clear identification and justification of the intended stakeholders and how the project relates to them	Stakeholders are listed but not clearly related to rest of project	Neither the stakeholders nor how the work relates to them are clearly defined
Initial conceptual model (*what is your initial idea for the design of the solution?*)	An initial, high level description of the planned solution which clearly identifies the team's current intentions for the design. Thought has been given to how this relates to background work and results from any study carried out so far.	Idea is adequately defined, but not well justified or related to other work in the project.	No coherent description or justification of the idea
Questions/areas of investigation (*what areas of knowledge have you identified that you need to investigate to learn more about the problem/ solution?*)	Insightful points raised that demonstrate a good grasp of the team's current understanding of the problem space; what further knowledge is required to improve understanding and advance the design process; and what activities will be required to acquire this knowledge.	Basic questions identified with some thought given to how to answer them	Little effort shown to identify questions or how to satisfy them

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
Design activities (*what are you doing to explore the solution space?*)	Relevant activities have been planned and/or carried out that are designed to learn more about the problem and/or ideas for solution.	Activities are described which show that the team is considering stakeholder perspectives and input on the nature of the problem and the building towards a solution.	Little or no activities have been planned or undertaken to seek any input into the design of the project from outside of the immediate knowledge of the team members themselves.The project is described in terms of development of 'a prototype'. Prototyping is considered solely in terms of a final product.
Ethical considerations *(How has the team addressed ethical concerns in relation to the topic of the project and the people who will be impacted by it?)*	Ethical considerations are central to the design process being followed and the team's understanding of ethics in relation to their project is a key driver for the development of their design. The team have actively used the ethical considerations for their project as a way to determine not only what they *should* be doing, but also what they *should not* do and why.	Ethical considerations have been listed and considered in an appropriate way for the project, but it is not clear how the team are incorporating them into the design.	Ethical concerns have not been identified or have been ignored. The project is progressing in a direction that may have a negative impact on the lives of the people who are being designed for
Project Plan (*how are you planning your work as a team to achieve the outcomes you are aiming for?*)	Clear breakdown of project into milestones, deliverables, strands, etc. as appropriate for the current stage of the design, and documentation of any agreed processes for team maintenance.	Basic description of project plan, with minimal consideration for how the team will work effectively towards achieving it.	Basic task breakdown only and little/no evidence of any discussion about how the team members will work together.

	EXCELLENT	SATISFACTORY	UNSATISFACTORY
COMMUNICATION (PRESENTATION) (weight 2) It is always important to remember when communicating about your work to others to be focused on how they will receive and understand your work. Due to the short time span for this assessment item, we will be mainly focusing on a satisfactory quality of work.			
Communication of Content (*how well did you explain what your project is about?*)	Clear coherent flow of concepts and appropriate level of detail, easy to follow and understand, clearly relevant design for purpose	Good concepts and use of images/text/ materials, but not well related to challenge	No clear flow of concepts, resulting in a confused message; content either too high or too low level
Audio/Visual Presentation (*how well did you present your work orally?*)	Holds audience attention and conveys ideas strongly	Clearly audible, good flow and audience contact	Unclear and forced, little engagement, just read out slide contents
Communication Design (*what attention have you given to the design of written and presented work?*)	The design of the presentation, visuals, etc. fit the context well and demonstrate flare in the team's approach to the problem	Presentation design appropriate for the content, but the balance of content between visual, textual and presentation of data could be improved.	Unclear how the design of the presentation contributed to content, or the design detracts from understanding the project
PROFESSIONALISM (WRITTEN REPORT) (weight 2) This report is essentially a first draft of what will become the final report for the studio. For a draft, the following criteria will not be applied, but they are included here to provide guidance on what will be looked at in more detail as we progress towards the end of semester.			
Language and Writing Style	Fluent and literate. Of high standard with good structure and flow	Adequate, but needs improvement	Of low standard, hard to understand and follow meaning
Referencing	Excellent support for the text and is of a consistent style throughout	Just adequate, or is not consistent style	Inadequate in number, or lack of consistent format
Proof Reading	No errors, clear that care has been taken to present a high quality report	Some errors have been missed, but well presented overall	Clearly not been checked for spelling, grammar, and typographical errors

Version information

Version 1.0	15/03/2023	Document created based on revision of the assessment task sheet from 2022.	S.Viller